



SYMBIOSIS INSTITUTE OF TECHNOLOGY
(Established under section 3 of the UGC Act 1956)
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ASSIGNMENT No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement :

Write a Java program to demonstrate the use of the super keyword in inheritance

Theory:

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Program :

```
class Animal
{
    String color="White";
    Animal()
    {
        System.out.println("Animal Constr");
    }
    void sound()
    {
```

```

        System.out.println("Animal SOunds");
    }
}
class Dog extends Animal
{
    String color="Black";
    Dog()
    {
        super();
        System.out.println("Dog Constr");
    }
    void display()
    {
        System.out.println("Dog color: "+color);
        System.out.println("Animal color: "+super.color);
        super.sound();
    }
}
}
public class SuperKeywordDemo
{
    public static void main(String[]args)
    {
        Dog d=new Dog();
        d.display();
    }
}

```

Output :

```
PS C:\Users\HP> cd C:\Users\HP\Documents\java
PS C:\Users\HP\Documents\java> javac SuperKeywordDem
PS C:\Users\HP\Documents\java> java SuperKeywordDemo
Animal Constr
Dog Constr
Dog color: Black
Animal color: White
Animal S0unds
PS C:\Users\HP\Documents\java> |
```

Exercise 1 Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Program :

```
class Employee {

    String name = "Sourabh Gupta";
    int salary = 50000;

    void displayEmployee() {
        System.out.println("Employee Name : " + name);
        System.out.println("Salary      : " + salary);
    }
}
```

```

class Manager extends Employee {

    String department = "Computer Science";

    void displayManager() {

        // Accessing Employee details using super
        System.out.println("Employee Name : " + super.name);
        System.out.println("Salary      : " + super.salary);

        System.out.println("Department   : " + department);
    }
}

public class EmployeeManager {

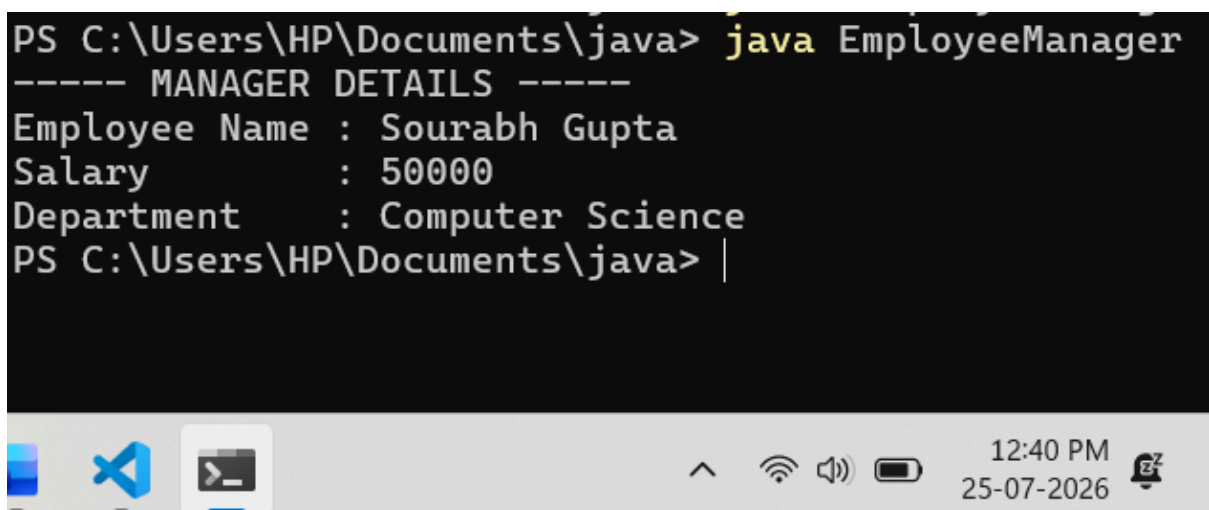
    public static void main(String[] args) {

        Manager m = new Manager();

        System.out.println("----- MANAGER DETAILS -----");
        m.displayManager();
    }
}

```

Output:



```

PS C:\Users\HP\Documents\java> java EmployeeManager
----- MANAGER DETAILS -----
Employee Name : Sourabh Gupta
Salary      : 50000
Department   : Computer Science
PS C:\Users\HP\Documents\java> |

```

Exercise 2. Develop a Vehicle Insurance System where child insurance

classes access parent vehicle information using the super keyword.

Source code:

```
class Vehicle {

    String vehicleNumber = "MH12AB1234";
    String vehicleModel = "Honda City";
    int vehicleYear = 2023;

    void displayVehicle() {
        System.out.println("Vehicle Number : " + vehicleNumber);
        System.out.println("Vehicle Model : " + vehicleModel);
        System.out.println("Vehicle Year : " + vehicleYear);
    }
}

class VehicleInsurance extends Vehicle {

    String insuranceType = "Comprehensive";
    double premium = 15000;

    void displayInsurance() {

        // Accessing parent class information using super
        System.out.println("Vehicle Number : " + super.vehicleNumber);
        System.out.println("Vehicle Model : " + super.vehicleModel);
        System.out.println("Vehicle Year : " + super.vehicleYear);

        System.out.println("Insurance Type : " + insuranceType);
        System.out.println("Premium : Rs. " + premium);
    }
}

public class VehicleInsuranceDemo {

    public static void main(String[] args) {

        VehicleInsurance insurance = new VehicleInsurance();

        System.out.println("----- VEHICLE INSURANCE DETAILS -----");
        insurance.displayInsurance();
    }
}
```

}

Output:

```
PS C:\Users\HP\Documents\java> java Vehicle
----- VEHICLE INSURANCE DETAILS -----
Vehicle Number : MH12AB1234
Vehicle Model   : Honda City
Vehicle Year    : 2023
Insurance Type  : Comprehensive
Premium        : Rs. 15000.0
PS C:\Users\HP\Documents\java> |
```

